Madhav Institute of Technology & Science, Gwalior
Department of Electronics & Communication Engineering

CORNERSTONE PROJECT
B.Tech · 2026

COLLABORATIVE CIRCUIT CAD & DC SIMULATION

CircuitCollab

Draft, share and simulate DC electronic circuits together — live, in any browser, with nothing to install.

Live Simulation · Real-time Collab · MNA DC Solver

0 installs needed

6 character room code

100% benchmark accuracy



Live editor: an LED circuit solved instantly, results table below the canvas.

01 The Problem

1. **Desktop isolation.** Reviewing a schematic means exporting files, sending screenshots or sharing screens.
2. **Installation friction.** Multi-gigabyte installs and licences slow down lab teaching.
3. **Unsynchronised simulation.** Remote teammates test different parameters and get different results.

02 Our Solution

1. **Zero-install web CAD.** The full schematic editor opens instantly in any modern browser.
2. **Google Docs-style rooms.** Join with a 6-character code and see live cursors, parts, wires and edits.
3. **Deterministic MNA solver.** One solver on client and server gives everyone identical electrical metrics.

03 System Architecture

Frontend (Browser) · React 18 + Tailwind
React Flow canvas · Zustand store · Client MNA solver

WebSockets · JSON
`circuit-update`
`run-simulation`

Backend (Server) · Node.js + Express
Socket.IO engine · Room manager · Server MNA solver

04 Key Features

- **7 parts:** R, C, L, DC source, ground, LED, switch
- **CAD tools:** rotate, wire, undo/redo, grid snap, zoom
- **Live cursors,** one colour per collaborator
- **Results:** node voltages, current, ΔV and power
- **Clear errors:** missing ground, floating pins, shorts

05 Modified Nodal Analysis

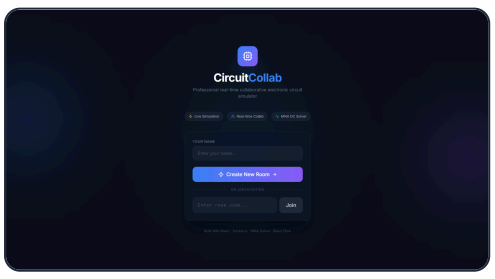
$$\begin{bmatrix} G & B \\ C & D \end{bmatrix} \begin{bmatrix} v \\ j \end{bmatrix} = \begin{bmatrix} i \\ e \end{bmatrix}$$

G conductances · **B, C** voltage-source incidence · **v** node voltages · **j** source currents · **e** source voltages

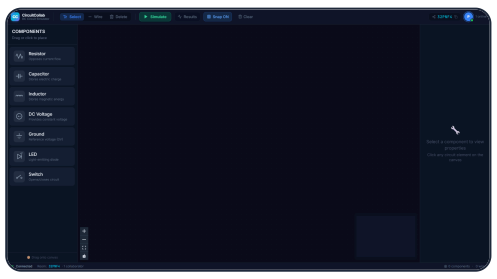
Gaussian elimination → Partial pivoting → Back-substitution

Singular matrix (pivot < 10^{-12}) is caught as a short circuit. LEDs use a piecewise-linear model with a ≈2.0 V threshold.

06 Inside the App



Lobby: create or join a room



Workspace: palette, tools, properties

07 Verified Results

Benchmark: 10 V source, 2 kΩ resistor. Hand check: I = 10/2000 = 5 mA, P = 50 mW.

Part	ΔV	I	P	Accuracy
V1 source	10.00 V	5.00 mA	50.00 mW	100.0%
R1 resistor	10.00 V	5.00 mA	50.00 mW	100.0%
GND	0.00 V	—	—	Reference

08 Tech Stack

React 18 · TypeScript · Vite · React Flow · Zustand · Tailwind CSS · Node.js · Express · Socket.IO

09 Future Scope

- AC & transient analysis
- BJT & Op-Amp models
- Virtual DMM & oscilloscope
- SPICE netlist & BOM export

PROJECT TEAM

Under the guidance of **Dr. Prateek Kumar**

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